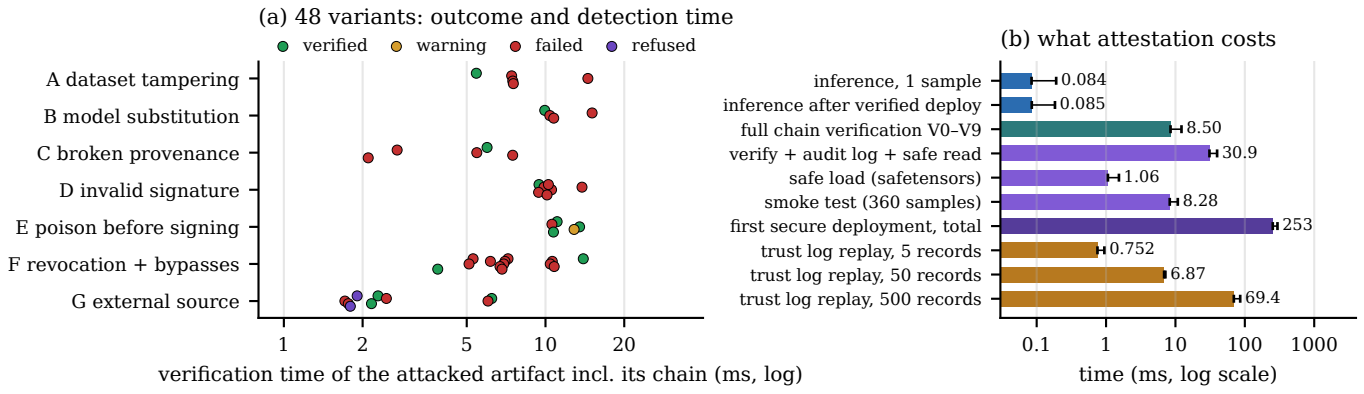
(a) 48 variants: outcome and detection time
verified
warning
failed
refused
A dataset tampering
B model substitution
C broken provenance
D invalid signature
E poison before signing
F revocation + bypasses
G external source
1
2
5
10
20
verification time of the attacked artifact incl. its chain (ms, log)
(b) what attestation costs
inference, 1 sample
0.084
inference after verified deploy
0.085
full chain verification V0–V9
8.50
verify + audit log + safe read
30.9
safe load (safetensors)
1.06
smoke test (360 samples)
8.28
first secure deployment, total
253
trust log replay, 5 records
0.752
trust log replay, 50 records
6.87
trust log replay, 500 records
69.4
0.1
1
10
100
1000
time (ms, log scale)